

Name:

ID:

Section: 05

- Q.1 Briefly explain the bus connection of a microcomputer system with suitable examples. **[5]** **[CLO1]**
- Q.2 A program running on the 8086 microprocessor accesses a memory location with a physical address of 3965Ch. **[5]** **[CLO2]**
- (a) Find the offset address if the segment number is 23FEh.
- (b) Find the segment number if the offset address is 164Ch.

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Course Code: CSE 313/ 07143119, **Course Title:** Microprocessor, Microcontroller and
Assembly Language Programming

CT-01 Fall - 2025 Full marks: 10 Time: 25 minutes

Name:

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Section: 05

- Q.1 Briefly explain how the CPU executes a machine instruction, describing the fetch and execute cycles with suitable examples. [5] [CLO1]
- Q.2 During memory addressing in an 8086 microprocessor, a program accesses a location with the physical address 4A65Ch. [5] [CLO2]
- (a) Find the offset address if the segment number is 31CEh.
- (b) Find the segment number if the offset address is 135Ch.

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Course Code: CSE 313/ 07143119, **Course Title:** Microprocessor, Microcontroller and
Assembly Language Programming

CT-02 Fall- 2025 Full marks 10 Time: 25 minutes

Name:

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Section: 05

- Q.1 Explain the startup operation of the 8086 microprocessor. [4] [CLO1]
- Q.2 For each of the following 8086 instructions, give the new destination contents. [6] [CLO2]
1. NEG AX where AX contains 001h.
 2. XCHG AX, BX where AX contains 135Ch and BX contains 783Dh.
 3. INC AL where AL contains FFh.

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Course Code: CSE 313/ 07143119, **Course Title:** Microprocessor, Microcontroller and
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CT-02 Fall- 2025 Full marks 10 Time: 25 minutes

Name:

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Section: 03

- Q.1 Briefly explain the Instruction Pointer (IP) and Flag Register of the 8086 microprocessor with suitable examples. [4] [CLO1]
- Q.2 Translate the following high-level language statements into equivalent Assembly Language instructions. [6] [CLO2]

I. $C = B - 2 \times A$

II. $B = -A + 5$